Data Engineering in Large Language Model Pretraining: A Survey

Yudong Li†, Ruiqi Zhou†, Peiru Yang, Shiyi Zheng, Linlin Shen, Xintian Li, Yongfeng Huang

***Abstract*—The growth of large language models (LLMs) has increased demand for high-quality pretraining text, motivating greater attention to the efficiency with which training data is converted into model capabilities. This survey examines data engineering for text-based LLM pretraining, with public technical reports and open corpus projects as its principal evidence. Following the progression from corpus construction to data utilization and evaluation, we review practices supported by large-scale training and distinguish them from methods whose transfer to frontier models remains uncertain. A common mathematical formulation connects document processing, sampling distributions, token-level objectives, and evaluation under resource constraints. We synthesize recurring practices and disagreements concerning deduplication, filtering, synthetic augmentation, mixture optimization, and staged training. Because public disclosure is incomplete and experimental protocols differ, the conclusions are qualified by their evidence and evaluation settings. The survey provides a reference for interpreting reported practices and identifying unresolved questions in pretraining data engineering.**

***Index Terms*—Data engineering, large language models, pretraining**

I. Introduction

Large language models (LLMs) acquire language representations and generative capabilities through pretraining on large text corpora. Empirical scaling laws relate improvements in predictive performance to model size, training data, and computation [1, 2]. The associated demand for training tokens has grown substantially. GPT-3 [3] used approximately 300B tokens, whereas DeepSeek V3 [4] and LLaMA 3 [5] report 14.8T and 15.6T tokens, respectively. Qwen 3 [6], GLM-5 [7], and DeepSeek V4 [8] report budgets in the tens of trillions. These figures describe reported training consumption rather than a uniform measure of unique corpus size, and their comparison must account for repetition, tokenization, and training stage.

Continued expansion faces constraints in both data supply and data utilization. Web corpora contain duplicated pages, extraction artifacts, and heterogeneous content, so raw volume does not directly measure usable training information. Villalobos et al. [9] project that demand may approach the available stock of human-generated public text between 2026 and 2032 under their assumptions; this is a conditional forecast rather than an established exhaustion date. Longpre et al. [10] document increasing access restrictions among high-quality web sources, while provenance, consent, and copyright complicate the use of otherwise accessible material [11]. These constraints motivate methods that improve learning from a given pool of admissible data.

The efficiency of current learning procedures provides a second motivation. Human learners acquire substantial linguistic competence from less text than contemporary LLMs, although the comparison is affected by differences in sensory input, interaction, and learning objectives [12]. This observation does not establish a direct efficiency bound for language models, but it motivates studying how training examples contribute to knowledge acquisition. Data engineering addresses this question through acquisition, cleaning, selection, mixing, and scheduling. RefinedWeb [13] demonstrates the effectiveness of processed web data relative to curated mixtures in its evaluation setting. FineWeb-Edu [14] and DCLM [15] further show that filtering choices can materially affect performance under controlled training budgets. Together, these results support treating data processing as an optimization variable alongside architecture and compute.

Existing surveys approach this subject from complementary perspectives. Albalak et al. [16] review data selection, while Wang et al. [17] examine data management across pretraining and supervised fine-tuning. Related work surveys the interaction between LLMs and data management [18], and efficiency methods spanning models, data, and training systems [19]. This survey retains a narrower focus on text-based pretraining and on practices supported by large-scale training. Public reports for GPT, LLaMA, Qwen, DeepSeek, Kimi, GLM, OLMo, and Phi, together with open corpus projects, form the principal evidence base. We organize these observations into a common pipeline, compare choices within each stage, and identify promising methods whose adoption or effectiveness at frontier scale remains unresolved.

The distinction between reported practice and general validity is central to this survey. A technical report can establish that a method was used without isolating its causal contribution. Conversely, omission from a report does not establish non-use. We therefore distinguish recurring practices, results obtained under specific experimental conditions, and open research questions. The disclosure inventory in Appendix A retains contextual evidence from broader model reports, but post-training and multimodal disclosures are not treated as evidence for text-only pretraining.

The remainder follows the original progression from model foundations to data construction, utilization, and evaluation. Section II introduces LLM architectures and formalizes pretraining. Section III reviews corpus construction, and Section IV examines how a corpus is converted into training

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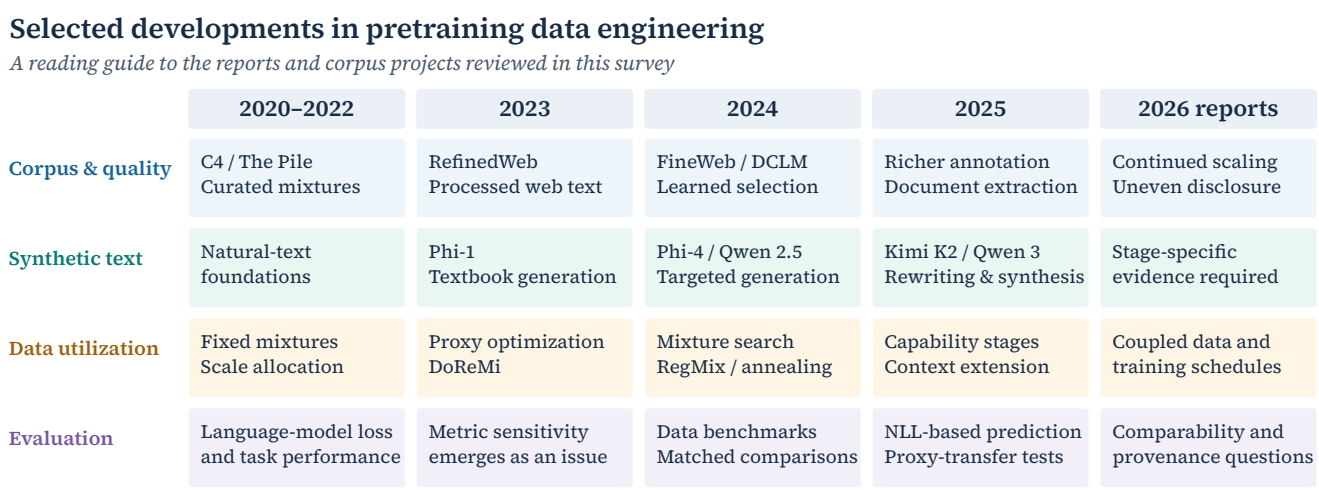


Fig. 1. Selected developments in pretraining data engineering. The timeline summarizes practices discussed in this survey; it does not imply uniform adoption or comparable training budgets.

signals. Section V discusses evaluation and comparability. Section VI synthesizes the shared pipeline and empirical consensus, and Section VII concludes with future directions.

II. Preliminaries

This section compares the principal Transformer architectures (§II-A) and formalizes the pretraining process (§II-B). The resulting notation connects the construction, utilization, and evaluation of training data throughout the survey.

A. LLM Architecture

Attention and Transformer blocks. Transformer language models process token representations through attention and position-wise transformations. Let $H \in \mathbb{R}^{L\times d}$ contain the hidden representations of a length-L sequence. For one attention head, queries, keys, and values are $Q = HW_Q$, $K = HW_K$, and $V_h = HW_V$. Attention is computed as

$$\mathrm{Attn}(H; M) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right) V_h, \tag{1}$$

where d_k is the key dimension and the softmax is applied row-wise. The mask M_{ij} equals 0 for permitted interactions and $-\infty$ otherwise. Multiple heads capture different interactions, and their outputs are combined by a learned projection. Positional representations, normalization, feed-forward transformations, and residual connections complete the block. For example, a pre-normalized block can be written as

$$\widetilde{H} = H + \mathrm{MHA}(\mathrm{Norm}(H); M), \tag{2}$$

$$H' = \widetilde{H} + \mathrm{FFN}(\mathrm{Norm}(\widetilde{H})). \tag{3}$$

These equations specify a representative computational structure rather than a universal ordering of all architectural components.

Architectural families. The attention pattern and prediction objective distinguish three architectural families (Figure 2). *Encoder-only* models permit bidirectional attention over the input and learn contextual representations, commonly through masked-token prediction. *Encoder–decoder* models first encode a source sequence bidirectionally and then generate a target sequence with causal self-attention and source cross-attention. T5 exemplifies this family through text-to-text denoising [20]. *Decoder-only* models use causal attention, with $M_{ij} = 0$ if $j \le i$ and $-\infty$ otherwise, to generate text from a prefix. GPT-3, PaLM, and LLaMA illustrate the use of this family for large-scale generative pretraining [3, 21, 22].

Architecture and learning objectives. The three families can consume the same raw text but expose different training targets. For a masked position set $\mathcal{M}$ and corrupted input $\widetilde{x}$, a representative masked-language-model objective is

$$\mathcal{L}_{\mathrm{MLM}} = -\mathbb{E}_{x,\mathcal{M}}\left[\frac{1}{|\mathcal{M}|}\sum_{t\in\mathcal{M}} \log p_\theta(x_t \mid \widetilde{x})\right]. \tag{4}$$

For encoder–decoder denoising, a corruption process produces a source u and a reconstruction target y, giving

$$\mathcal{L}_{\mathrm{denoise}} = -\mathbb{E}_{(u,y)}\left[\frac{1}{|y|}\sum_{t=1}^{|y|} \log p_\theta(y_t \mid y_{<t}, u)\right]. \tag{5}$$

The corruption scheme determines whether y contains the original text or selected spans. Decoder-only pretraining instead uses the next-token objective defined below. Architecture and objective are related design choices, but they are not interchangeable definitions: sequence transformations can also provide a causal decoder with infilling targets.

Dense and sparse decoders. Within the decoder-only family, substantial architectural variation remains. LLaMA combines RMS normalization, gated feed-forward layers, and rotary positional representations [5, 22]. Grouped-query attention reduces the number of key/value heads relative to query heads, while sparse mixture-of-experts (MoE) layers activate only a subset of feed-forward experts per token [4, 6, 23]. A representative MoE layer has the form

$$\mathrm{MoE}(h) = \sum_{e\in \mathrm{TopK}(r(h))} g_e(h) f_e(h), \tag{6}$$

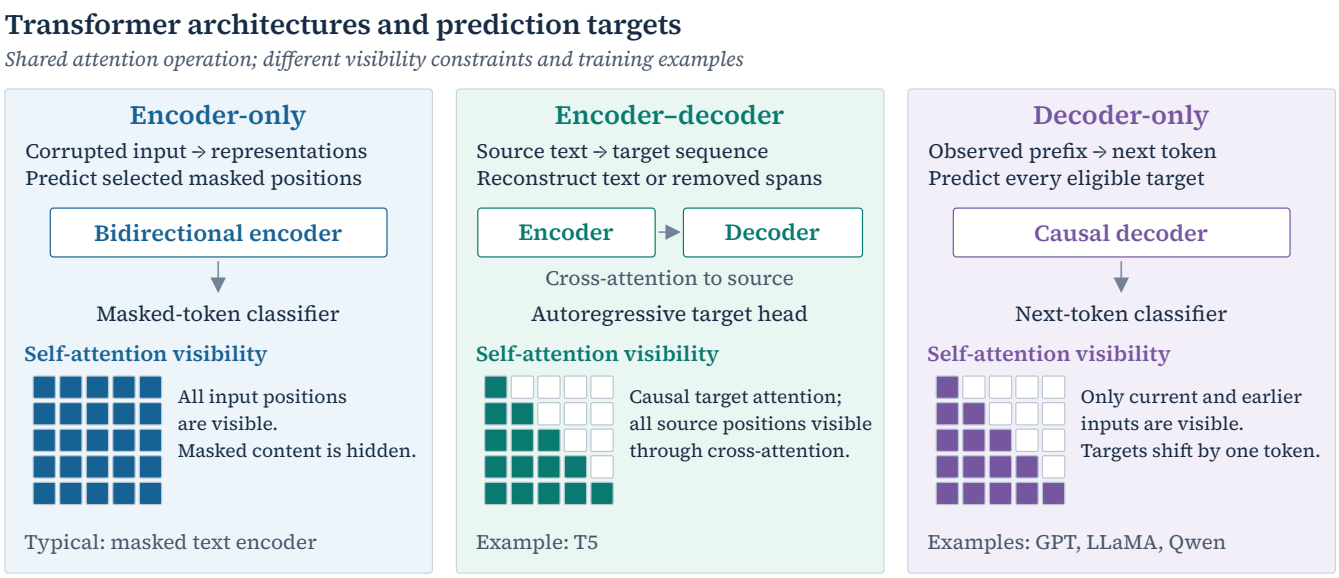


Fig. 2. Transformer architecture families and their attention patterns. Filled cells indicate permitted attention (rows are queries; columns are keys). Encoder–decoder models additionally use cross-attention from target positions to source representations.

where r is a router, f_e is expert e, and g_e is its routing weight. Total parameters P and active parameters P_{act} should therefore be distinguished. Sparse experts change computation and optimization without necessarily changing the next-token target. The prevalence of causal decoders supports a common data-centric formulation, but it does not imply convergence to an identical architecture.

B. LLM Pretraining

Definition 1 (Corpus and training exposure)**.** *Let $\mathcal{R} = \{r_i\}_{i=1}^{n}$ be an indexed collection of raw documents. A construction procedure $\mathcal{C}$ produces a tokenized corpus $\mathcal{D} = \mathcal{C}(\mathcal{R}) = \{x_i\}_{i=1}^{m}$, where $x_i = \tau(r'_i)$, τ is a fixed tokenizer with vocabulary $\mathcal{V}$, and r'_i is a retained or transformed document. Let $U = \sum_{i=1}^{m} |x_i|$ denote the stored token count. The training exposure B counts token presentations consumed by training, including repeated presentations of the same stored tokens.*

Thus U is not a count of distinct token identities or semantic facts, and B need not equal U. We use K to denote the number of source or domain strata $\mathcal{D}_k$. When domains overlap, an explicit assignment or weighting rule is required before domain proportions can be interpreted. This distinction prevents corpus volume, training exposure, and domain coverage from being conflated.

Definition 2 (Data utilization policy)**.** *At optimization step s, a data utilization policy $\pi_s = (q_s, \mathcal{T}_s, L_s, w_s)$ specifies a distribution q_s over eligible documents, a sequence transformation $\mathcal{T}_s$, a maximum context length L_s, and nonnegative target weights w_s. A transformation may tokenize, crop, pack, or rearrange text; its attention and loss masks are part of the resulting training example.*

Here tokenization in $\mathcal{T}_s$ is needed only when sampling precedes tokenization. The learning rate η_s is an associated optimization choice, rather than a property of the corpus. Corpus construction changes the available examples; utilization changes their exposure, context, and contribution to the objective.

Autoregressive factorization. For a token sequence $x = (x_1, \ldots, x_L)$, an autoregressive model factors its likelihood as

$$p_\theta(x) = \prod_{t=1}^{L} p_\theta(x_t \mid x_{<t}), \tag{7}$$

where the empty prefix at $t = 1$ is represented by a beginning-of-sequence context. If h_{t-1} encodes that prefix, the output projection gives

$$p_\theta(x_t = v \mid x_{<t}) = \frac{\exp((W_o h_{t-1} + b)_v)}{\sum_{u \in \mathcal{V}} \exp((W_o h_{t-1} + b)_u)}. \tag{8}$$

Targets are shifted relative to inputs so that a token is never used to predict itself. During teacher-forced training the prefix contains observed tokens; during generation it contains previously generated tokens. The factorization is shared across training and inference, although their prefix distributions can differ.

Masked and weighted supervision. For a batch of transformed sequences $\mathcal{B}_s$, define $\ell_{i,t}(\theta) = -\log p_\theta(x_{i,t} \mid c_{i,t})$, where $c_{i,t}$ is the context permitted by the attention mask. The normalized training objective is

$$\begin{aligned} Z_s &= \sum_{i \in \mathcal{B}_s} \sum_t m_{i,t} w_{s,i,t}, \\ \widehat{\mathcal{L}}_s(\theta) &= \frac{1}{Z_s} \sum_{i \in \mathcal{B}_s} \sum_t m_{i,t} w_{s,i,t} \ell_{i,t}(\theta), \end{aligned} \tag{9}$$

assuming $Z_s > 0$. The binary mask $m_{i,t}$ removes padding and any positions excluded from supervision; $w_{s,i,t} = 1$ recovers ordinary token-averaged cross-entropy. Attention masks determine which tokens may be read, whereas loss masks determine

Algorithm 1 Pretraining under a data utilization policy

Require: Corpus $\mathcal{D}$, tokenizer τ, policies π_s, token budget B

Initialize model parameters θ and optimizer state
Set $s \leftarrow 1$ and consumed token presentations $b \leftarrow 0$
while $b < B$ **do**
  Sample under q_s within the remaining exposure budget
  Apply $\mathcal{T}_s$ with context bound L_s
  Construct shifted targets, attention masks, and $m_{i,t}$
  Evaluate Eq. (9) with weights w_s
  Update parameters using the optimizer and η_s
  Increase b by the actual non-padding token presentations
  Evaluate checkpoints and set $s \leftarrow s + 1$
end while
return Pretrained parameters θ

which targets contribute gradients. They are distinct even when implemented in the same batch pipeline.

Mini-batch optimization. A mini-batch algorithm samples documents under q_s, constructs contexts with $\mathcal{T}_s$, evaluates Eq. (9), and updates θ using its gradient. For completeness, a representative AdamW update is

$$\begin{aligned} g_s &= \nabla_\theta \widehat{\mathcal{L}}_s(\theta_s), \\ a_s &= \beta_1 a_{s-1} + (1-\beta_1) g_s, \\ v_s &= \beta_2 v_{s-1} + (1-\beta_2) g_s \odot g_s, \\ \theta_{s+1} &= (1-\eta_s\lambda)\theta_s - \eta_s \frac{\widehat{a}_s}{\sqrt{\widehat{v}_s} + \epsilon}, \end{aligned} \tag{10}$$

where $a_0 = v_0 = 0$, $\widehat{a}_s = a_s/(1-\beta_1^s)$, and $\widehat{v}_s = v_s/(1-\beta_2^s)$. All vector operations are element-wise. The coefficient λ denotes decoupled weight decay. Optimizer details vary across reports; the formulation identifies how data choices influence the gradient rather than prescribing one optimizer for every model.

Training exposure and computational cost. This formulation also clarifies resource comparisons. A frequently used approximation for dense Transformer training is $F \approx 6PB$, where F is the floating-point operation count, P is the parameter count, and B is training exposure [1, 2]. It omits sequence-length-dependent attention costs and other implementation overheads; using P_{act} for MoE models is only a rough extension. A reported improvement in tokens-to-target does not necessarily imply the same improvement in total compute, particularly when filtering or synthesis is expensive. We therefore distinguish exposure B, model-training compute F, and the additional cost of constructing $\mathcal{D}$.

III. Corpus Construction

Corpus construction specifies the transformation $\mathcal{D} = \mathcal{C}(\mathcal{R})$ in Definition 1. We review data curation (§III-A), deduplication (§III-B), quality filtering (§III-C), and synthetic augmentation (§III-D), retaining the distinction between available text and its eventual training exposure.

TABLE I
Notation used throughout the survey.

Symbol	Meaning
$\mathcal{R}, \mathcal{D}, \mathcal{D}_k$	Raw collection, constructed corpus, domain stratum
U, B, F	Stored tokens, training exposure, training compute
P, P_{act}	Total and active parameter counts
$q_s, \alpha_{s,k}$	Document distribution and domain sampling weight
$\mathcal{T}_s, L_s$	Sequence transformation and context bound
$m_{i,t}, w_{s,i,t}$	Binary loss mask and target weight
η_s, θ_s	Learning rate and model parameters at step s
$\mathcal{E}, \Gamma$	Evaluation data and evaluation protocol

A. Data Curation

Curated and web-scale corpora. Earlier public corpora often combined multiple curated sources. The Pile [24] aggregates 22 English sources totaling approximately 825 GiB, and CLUECorpus2020 [25] provides approximately 100 GB of Chinese text. C4 [20] establishes a reproducible approach to filtering CommonCrawl. Such collections broaden source coverage, but individually curated sources have limited capacity to supply the token volumes required by later models. Repeated CommonCrawl snapshots provide a large public reservoir of web text, including substantial overlap across pages and crawl dates. Their scale makes extraction, normalization, and provenance tracking integral parts of corpus construction.

Processed web data can be competitive with curated mixtures under controlled comparisons. RefinedWeb [13] demonstrates this result using a web-focused pipeline. FineWeb [14], DCLM [15], Dolma [26], and Nemotron-CC [27] subsequently provide different approaches to extraction, filtering, and corpus release. These projects support the use of web text as a major pretraining source without establishing that one source is sufficient for every capability. Table II summarizes their reported scales and evaluation settings. In particular, candidate-pool size, released corpus size, and model-training exposure are distinct quantities.

Domain-specific sources. Domain sources complement web text with code repositories, scientific papers, books, and encyclopedias. Their value depends on content and provenance rather than source labels alone: a scientific article may be peer-reviewed, whereas a repository or forum response may not be. Code, mathematical exposition, and long-form text provide structures that can be underrepresented in a generic web sample. Building these domain strata $\mathcal{D}_k$ still requires extraction, normalization, deduplication, and quality control. Open corpus projects reduce the effort required to reproduce this processing and to study its effect on training.

Document extraction. PDFs and scans provide another route to scientific and long-form text, but their layout must be converted into reliable sequences. Formulas, tables, multiple columns, and reading order are recurrent extraction difficulties. Nougat [30] uses an end-to-end Transformer to transcribe academic documents, including mathematical structure. MinerU [31] addresses a wider range of digital documents and scans, while olmOCR [32] uses a vision-language model for

TABLE II
MAJOR OPEN PRETRAINING CORPORA. PERFORMANCE NUMBERS ARE COLOR-CODED BY EVALUATION SETUP: **FW-EVAL** = AGGREGATE OF 8 NLP BENCHMARKS, 1.8B MODEL, 350B TOKENS [14]; **DCLM-EVAL** = MMLU, 7B MODEL, 280B TOKENS [15]; **NC-EVAL** = AVERAGE OF 20 BENCHMARKS, 8B MODEL [27]. COMPARISONS REQUIRE THE SAME PROTOCOL WITHIN EACH GROUP; SCORES ACROSS GROUPS ARE NOT DIRECTLY COMPARABLE. SCALE DENOTES THE REPORTED CORPUS OR CANDIDATE-POOL SIZE, NOT A COMMON TRAINING BUDGET. DCLM 240T DENOTES ITS CANDIDATE POOL, NOT THE FILTERED TRAINING SET.

Corpus	Scale	Design Focus	Performance
C4 [20]	156B	Rule-filtered CommonCrawl corpus	FW-Eval 48.2
The Pile [24]	825 GiB	Add data sources to broaden general knowledge	FW-Eval 46.8
RefinedWeb [13]	5T	Web-only data can match curated multi-source corpora	FW-Eval 48.8
SlimPajama [28]	627B	Global deduplication across sources	FW-Eval 46.0
Dolma [26]	3T	Fully open and reproducible pipeline	FW-Eval 47.5
RedPajama V2 [29]	30T	Flexible quality filtering with 40+ quality signals	FW-Eval 46.7
FineWeb [14]	15T	Stage-by-stage ablations of the data pipeline	FW-Eval 48.8
FineWeb-Edu [14]	1.3T	Filtering targeted at educational content	FW-Eval **50.5**; DCLM-Eval 37.3; NC-Eval 53.2
DCLM [15]	240T	Standardized data-selection benchmark	DCLM-Eval **50.8**; NC-Eval 57.0
Nemotron-CC [27]	6.5T	Classifier ensembling and synthetic rewriting	NC-Eval **57.8**

page-level transcription. OLMo 3 [33] incorporates extracted academic and book content and uses synthetic question–answer data in mid-training. Although visual models may perform extraction, the resulting training examples remain text, consistent with this survey's scope. Extraction quality should be evaluated separately from the informational value of the source document.

B. Deduplication

Deduplication controls accidental repetition in the stored corpus. Web redundancy includes identical or near-identical content across URLs and repeated observations of the same page across crawl snapshots. These mechanisms differ from deliberate resampling during training. The former changes $\mathcal{D}$ and its implicit source weights; the latter changes the exposure distribution q_s while leaving the stored corpus fixed. This distinction is necessary when comparing claims about the benefits and harms of repetition.

Definition 3 (Lexical near-duplication)**.** *For a tokenized document x, let $G_n(x)$ be its set of n-grams. A pair of nonempty shingle sets has Jaccard similarity*

$$J(x, x') = \frac{|G_n(x) \cap G_n(x')|}{|G_n(x) \cup G_n(x')|}. \tag{11}$$

A threshold $\delta \in [0, 1]$ identifies candidate near-duplicates when $J(x, x') \geq \delta$. The threshold relation need not be transitive, so clustering and representative selection must be specified separately.

The operational definition concerns lexical overlap rather than semantic equivalence. Exact deduplication compares normalized content hashes; URL matching instead identifies repeated addresses and can miss content changes at a fixed address. OLMo 1 [34] and the Dolma pipeline [26] use Bloom-filter-based processing to reduce the memory required for duplicate detection. Bloom filters can return false positives, so the false-positive rate is part of the retention policy.

Scalable near-duplicate detection. MinHash estimates Eq. (11), and locality-sensitive hashing (LSH) retrieves likely matches without comparing every document pair. For an ideal minwise hash h, $\Pr[h(G_n(x)) = h(G_n(x'))] = J(x, x')$. Multiple signatures reduce estimation variance, while LSH trades candidate recall against comparison cost. At a fixed number of signatures, storage remains approximately linear in the number of documents. Its practical advantage is compact representations and fewer candidate comparisons, not a general guarantee of sublinear memory. Performance also depends on bucket skew, shingle definitions, and the treatment of short documents.

Granularity and corpus coverage. Reported pipelines differ in granularity and scope. LLaMA 3 combines URL, document-level, and line-level deduplication to address both replicated documents and recurring fragments. DeepSeek V2 [35] describes cross-snapshot processing to control temporal redundancy, which remains relevant to later releases. InternLM 2 [36] reports MinHash settings including 128 hash functions, 5-grams, and a Jaccard threshold of 0.7. It also describes code-specific organization to preserve useful repository structure. Such details are reproducible configuration choices rather than universal defaults. Cross-source and cross-snapshot coverage can matter as much as the matching algorithm.

Coverage and repetition trade-offs. Aggressive removal can reduce useful coverage when independently written documents contain shared explanations or common facts. High

overlap does not imply that all copies have identical training value, and common knowledge may be expressed with repeated terminology. Thresholds and representative selection should therefore be assessed jointly with retained domain coverage. Conversely, a high observed frequency can reflect scraping artifacts or spam rather than importance. Deduplication alone cannot reliably infer the educational or factual value of a repeated passage.

Empirical results on repetition depend on the source of redundancy and the data budget. Hernandez et al. [37] show that concentrated repetition can reduce effective model capacity in their scaling experiments. Carlini et al. [38] associate repeated training sequences with increased verbatim memorization. These findings support controlling unintended repetition, but do not imply that every additional epoch is harmful. Muennighoff et al. [39] show that repeated training can remain beneficial when fresh data is limited, with diminishing returns. The relevant comparison is therefore between alternative uses of a fixed budget B, including access to fresh text, rather than between repetition and non-repetition in isolation.

C. Quality Filtering

Quality filtering selects documents according to an operational criterion. Let $s_\phi(x)$ be a scoring function and τ_q a threshold. A deterministic filter produces

$$\mathcal{D}_{\text{keep}} = \{x \in \mathcal{D} : s_\phi(x) \geq \tau_q\}. \tag{12}$$

Soft selection instead modifies the probability of sampling a document. Neither formulation supplies a task-independent definition of quality: the labels, reference distribution, and downstream evaluation determine what the score rewards. Retention rate and the domain composition after filtering are therefore necessary complements to aggregate benchmark gains.

Heuristic and perplexity filters. Practical pipelines often combine inexpensive rules with learned scoring. The Gopher rules [40] threshold document length, word statistics, and repetition-related features to remove obvious extraction artifacts and low-information text. Perplexity-based filtering measures how well a reference language model predicts each document. Yi [41] and Falcon [42] report such filtering, although a high perplexity may indicate domain novelty as well as corruption. Rules and reference-model scores are consequently useful as coarse signals whose thresholds require validation.

Learned quality criteria. Learned filters depend strongly on how positive examples are defined. GPT-3 uses reference sources including Reddit-linked content, and LLaMA 1 [22] uses Wikipedia-related quality signals in its CommonCrawl processing. LLaMA 3 distills LLaMA 2-generated quality judgments into a smaller classifier. DCLM [15] studies fastText-based selection with curated positive data, including instruction-style examples, and reports a 7B model reaching approximately 64% on MMLU. Its reported compute comparison with LLaMA 3 is an end-to-end model comparison, not an isolated causal estimate of the filter. Controlled comparisons within the DCLM framework provide the stronger basis for attributing gains to selection.

Distilling model judgments. FineWeb-Edu [14] uses Llama-3-70B-Instruct to score 500K examples for educational value and trains a lightweight classifier to scale the selection procedure. Nemotron-CC [27] combines complementary quality signals and synthetic transformations, while Qwen 3 describes multidimensional annotation and data selection. These systems separate expensive judgment generation from lower-cost corpus-wide scoring. The resulting labels can inform both Eq. (12) and the domain weights used during training. Their computational advantage depends on annotation volume, classifier cost, and the size of the candidate corpus.

Selection bias. The principal limitation is selection bias. A classifier trained on encyclopedic or educational prose may reject informal documents that still contain useful information. DeepSeek's discussion of relaxed filtering in V2 illustrates the possibility that strict selection reduces performance on knowledge-intensive tasks. Yi's emphasis on a smaller, more selective corpus illustrates a different operating point. Neither observation establishes a universally optimal threshold. A filtering decision should be interpreted relative to the retained distribution, training exposure, and capabilities used for validation.

D. Synthetic Data Augmentation

Synthetic augmentation uses a model to generate or transform text. Let q_{nat} and q_{syn} be distributions over natural and accepted synthetic training examples. Their mixture is

$$q_\rho = (1-\rho)q_{\text{nat}} + \rho q_{\text{syn}}, \qquad 0 \leq \rho \leq 1. \tag{13}$$

Here ρ is an example-sampling probability; its token share also depends on example lengths. Synthetic generation may be grounded in a source document or conditioned only on a prompt and the generator's parameters. The provenance of seeds, validation rules, and mixing convention are necessary to distinguish these settings.

Generative augmentation. Reported use varies across model families. DeepSeek V3's description of web and ebook data does not by itself establish that every pretraining example is free of model-generated content. Phi-4 [43], in contrast, explicitly assigns a substantial role to synthetic data, reporting a 40% synthetic share in its pretraining mixture. Phi-1 [44] demonstrates the use of generated textbook-style material for code learning, and later Phi models expand the approach. Phi-4 reports more than 50 synthetic dataset categories with approximately 400B stored synthetic tokens. That stored volume and the synthetic share of training exposure are different quantities. Transfer of this recipe beyond the scales studied in the Phi series requires further evidence.

Other models use synthesis to strengthen particular domains. Qwen 2.5 [45] employs specialized models for mathematical and code data, and Qwen 3 extends generation to a broader range of subjects and formats. Related practices in InternLM use a preceding model generation to support subsequent corpus construction. OLMo 3 [33] combines document extraction with question–answer generation for mid-training. These examples show how synthesis can supply targeted contexts and supervision while retaining a natural-text foundation. They do not establish a universal synthetic-data ratio.

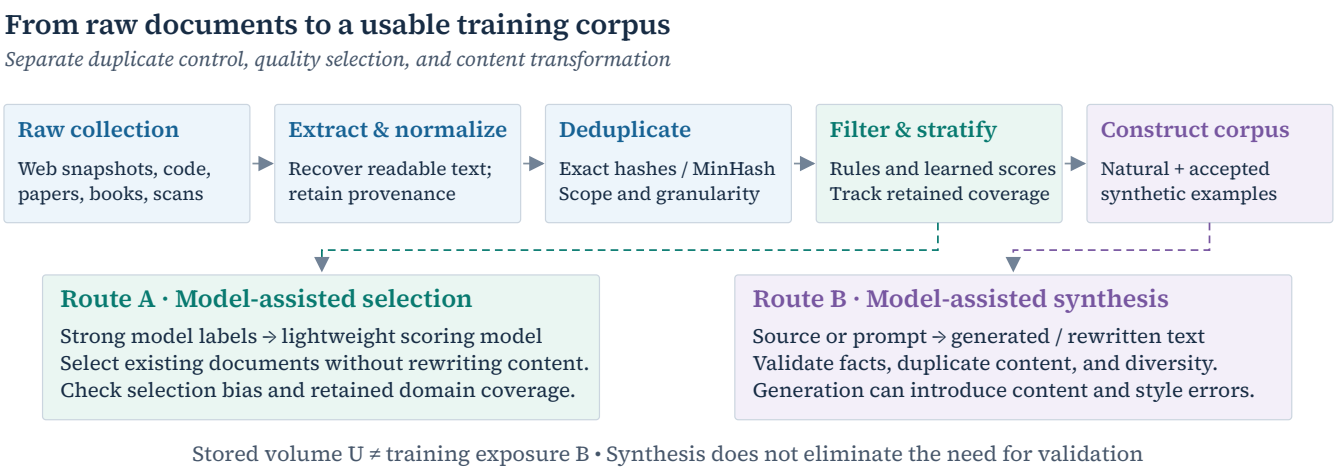


Fig. 3. Corpus construction and model-assisted processing. Filtering selects existing examples; synthesis changes their content. Both affect the retained distribution and require validation before sampling for training.

Source-conditioned rewriting. Source-conditioned rewriting provides a complementary approach. Kimi K2 [46] rewrites documents into different styles and perspectives, including mathematical notes and translated variants, and limits the number of rewrites per source. Its SimpleQA comparison supports rewriting over straightforward repetition in the reported setting. Nemotron-CC describes approximately 1.9T generated tokens, with transformations chosen according to source quality, including denoising, question–answer pairs, summaries, and knowledge lists. Grounding a rewrite in a source can constrain its content, but cannot guarantee factual preservation or eliminate the generator's stylistic and distributional biases. Validation must therefore address both correctness and diversity.

Mixture sensitivity and recursive risks. Controlled experiments further show that benefits depend on the natural–synthetic mixture. Kang et al. [47] report improvements in training efficiency for selected mixed-data settings, including a one-third synthetic mixture, while purely synthetic alternatives do not consistently outperform natural web data. The reported 5–10× gains are specific to their models, tasks, and efficiency definitions. Recursive replacement of natural data can also cause distributional degradation under the conditions studied in model-collapse research [48]. These findings motivate preserving natural-data coverage and evaluating successive synthesis rounds, rather than treating either collapse or efficiency improvement as an inevitable outcome of using synthetic text.

IV. Data Utilization

Data utilization determines how the corpus $\mathcal{D}$ contributes to parameter updates through the policy π_s. We review selection and mixing (§IV-A), multi-stage training (§IV-B), long-context training (§IV-C), and fine-grained training signals (§IV-D).

A. Data Selection and Mixing

For K domain strata, a mixture can be represented as

$$q_s(x) = \sum_{k=1}^{K} \alpha_{s,k} q_k(x), \quad \alpha_{s,k} \geq 0, \quad \sum_{k=1}^{K} \alpha_{s,k} = 1, \tag{14}$$

where q_k specifies sampling within stratum $\mathcal{D}_k$. If $\alpha_{s,k}$ denotes document-sampling mass and the expected transformed length in domain k is $\bar{L}_{s,k}$, the corresponding expected token share is

$$\widetilde{\alpha}_{s,k} = \frac{\alpha_{s,k} \bar{L}_{s,k}}{\sum_j \alpha_{s,j} \bar{L}_{s,j}}. \tag{15}$$

Thus document and token mixing ratios coincide only under appropriate length normalization. Reported percentages should be interpreted in their original sampling convention.

Fixed mixture weights. Early models commonly fixed mixture weights using source availability and development experiments. LLaMA 1 reports a mixture including CommonCrawl at 67%, C4 at 15%, and GitHub at 4.5%. BLOOM [49], Falcon [42], and PaLM [21] also describe predetermined source or language mixtures. PaLM 2 [50] increases emphasis on multilingual and code data, whereas OLMo 1 uses the Dolma recipe without an extensive mixture search. Fixed ratios simplify implementation and reproduction, but choosing them remains an empirical problem. Their adequacy depends on the target model, budget, and desired capability profile.

Proxy-based mixture optimization. Proxy-based optimization reduces the cost of exploring Eq. (14). A generic formulation selects

$$\alpha^* \in \arg\min_{\alpha \in \Delta^{K-1}} \widehat{\mathcal{R}}_{\mathcal{E},\Gamma}(\alpha; P, B), \tag{16}$$

where Δ^{K-1} is the probability simplex and $\widehat{\mathcal{R}}$ is a surrogate for evaluation risk on data $\mathcal{E}$ under protocol Γ. LLaMA 3 reports scaling-law experiments to refine its mixture and subsequent adjustments informed by capability measurements. Qwen 2.5, Qwen 3, and DeepSeek V3 describe related data-allocation procedures. These disclosures support the use of empirical optimization, without demonstrating adoption of a particular academic algorithm.

Academic methods differ in the surrogate and search procedure. DoReMi [51] uses a proxy model and distributionally robust optimization based on domain excess loss to estimate weights. Data Mixing Laws [52] fits an analytic relationship between mixtures and loss, reporting training-step savings in

its experiments. RegMix [53] learns a regression predictor from small runs and reports lower search cost than its DoReMi baseline. BiMix [54] models the joint effect of mixture and training volume. UtiliMax [55] uses utility estimates and allocation constraints to formulate mixture design as portfolio optimization. Their reported savings refer to different search and training settings, so they should not be combined into a single efficiency ranking.

Transfer across training scales. The main assumption is that proxy results remain informative at the target scale. DataDecide [56] studies more than 30K checkpoints and finds substantial, but imperfect, transfer of data rankings. AutoScale [57] identifies settings where small-scale optima do not transfer, and Mizrahi et al. [58] show that preferred filtering thresholds can vary with model size. These findings are compatible: a proxy can be useful on average while failing for a particular decision. Target-scale validation, intermediate-scale checks, and sensitivity analysis are therefore needed when feasible. Equation (16) should be interpreted as conditional on P, B, and the evaluation protocol.

B. Multi-stage Training

Multi-stage training changes the utilization policy over the course of optimization. A piecewise-constant data schedule can be written as

$$\pi_s = \pi^{(j)} \quad \text{for } s \in I_j, \qquad j = 1, \ldots, J, \tag{17}$$

where I_j partitions the optimization steps into stages. Each stage may change domain weights, sequence transformations, context length, or target weights. The learning-rate schedule η_s can change independently. This formulation distinguishes a switch in data from a switch in optimization dynamics and permits their interaction to be studied.

Capability-focused stages. Several reports allocate most exposure to broad pretraining before emphasizing selected capabilities. Qwen 3 describes an initial stage with more than 30T tokens spanning 119 languages and dialects, a subsequent approximately 5T-token stage emphasizing science, code, and reasoning, and a final long-context stage. GLM-5 and DeepSeek V3 also report staged training, although their boundaries and objectives differ. LLaMA 3 adjusts data allocation together with optimization settings during training. These examples support a recurring design pattern rather than a universal stage count or token allocation.

Scope of mid-training. *Mid-training* usually denotes continued training between broad pretraining and later adaptation, but reports use the term inconsistently. OLMo 2 [59] and OLMo 3 [33] emphasize higher-quality mathematical, code, and question–answer data in later training. InternLM 2 [36] also discusses capability-oriented continued training; this reference should not be treated as a separate report for a later model version. Qwen 3's second stage has a related capability-focused role. In this survey, such a stage belongs to pretraining only when the relevant evidence concerns language-model training on text. Supervised instruction tuning and reinforcement learning remain outside the methodological scope even if a model report calls them mid-training.

Annealing and data quality. Annealing denotes learning-rate decay and is often accompanied by an increase in selected high-quality data. LLaMA 3 reports a 24-percentage-point GSM8K improvement for its 8B model during its final annealing phase, while the gain differs at larger scales. Related late-stage selection is described for OLMo 3 and Phi-4. A decaying learning rate, such as one reported for Kimi K2, does not by itself establish a simultaneous quality-focused data switch. MiniCPM [60] studies the warmup–stable–decay (WSD) schedule. Luo et al. [61] analyze interactions between learning rate and data quality, providing one account of why a late data change may have different effects from the same change earlier in training. GLM-4.5 reports limitations of WSD in its setting, so WSD should not be presented as universally preferable.

Mid-training and annealing consequently describe different aspects of Eq. (17). The former identifies a stage by its position or purpose, whereas the latter identifies a learning-rate regime. A stage can satisfy both descriptions. Their effectiveness depends on the starting checkpoint, domain exposure, optimization settings, and evaluation targets. Apparent gains cannot be attributed solely to higher-quality data when several variables change together.

Staging versus difficulty curricula. Classical curriculum learning orders examples by a notion of difficulty, while capability-oriented staging primarily changes domains or target skills. Zhang et al. [62] report training-step reductions of 18–45% for the curriculum settings they study. Those results do not establish that the staged recipes in frontier reports implement a global difficulty ordering. A transferable definition of difficulty across heterogeneous text remains unresolved, and the reports reviewed here provide limited evidence for such an ordering. Figure 4 therefore represents policy changes without assigning every stage a universal difficulty level.

C. Long-context Training

Long-context training increases the context bound L_s and changes which dependencies can contribute to learning. It supports applications such as long-document analysis and repository-level code understanding, but a larger nominal window does not ensure effective use of distant information. LLaMA 3, DeepSeek V3, Qwen 3, and Kimi K2 describe context extension after much of their shorter-context training. The central design questions are how to schedule the extension and which examples provide useful long-range structure.

Progressive context extension. Staged extension limits the cost of applying long contexts throughout pretraining. DeepSeek V3 reports approximately 14.8T tokens in its initial 4K-context training and uses YaRN [63] in later extension to 32K and 128K. LLaMA 3 increases the context window from 8K to 128K through several stages. Similar phased approaches appear in Kimi K2. Positional adaptation, sequence length, and corpus composition are coupled choices, so context-length effects should be evaluated at both short and long ranges.

Long-document supply and packing. Long natural documents are less abundant than short web passages, and their source distribution differs. Books, papers, and code repositories supply extended structure but may concentrate exposure

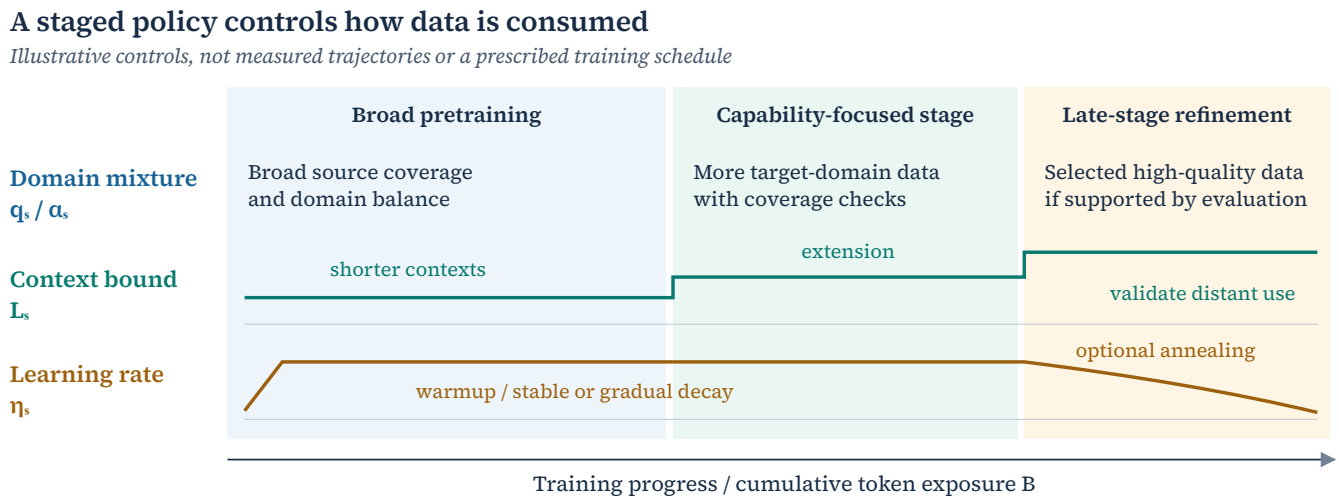


Fig. 4. A schematic staged utilization policy. The mixture, context bound, and learning rate are separate controls. Stage widths and curves are illustrative and do not represent measured token allocations or a universal recipe.

in a small number of domains. Upsampling these sources changes q_s, whereas packing changes $\mathcal{T}_s$. In-Context Pretraining [64] retrieves related documents for concatenation, aiming to provide more informative cross-document context. Best-fit Packing [65] addresses length utilization by reducing padding and truncation. These methods solve different problems: a nearly full training sequence is not necessarily a coherent long-context example.

Boundary-aware attention. Document boundaries require explicit attention and loss policies. For packed sequence positions with document identifiers $d(i)$, a document-isolated causal mask is

$$M_{ij} = \begin{cases} 0, & j \le i \text{ and } d(j) = d(i), \\ -\infty, & \text{otherwise.} \end{cases} \tag{18}$$

Allowing cross-document attention removes the second condition and changes the statistical training context. End-of-document tokens alone do not enforce attention isolation. Reports should also specify whether boundary predictions contribute to the loss. Synthetic retrieval or long-context examples, as described in later stages of Qwen 3 and other models, provide an additional way to create distant dependencies. Their benefit should be measured with held-out long-context tasks rather than inferred from the number of packed tokens.

D. Training Signals

Standard language modeling assigns equal target weights in Eq. (9), but token losses and gradients are not equal. Xia et al. [66] show that learning trajectories vary across tokens and model sizes. Chang et al. [67] observe learning, forgetting, and subsequent stabilization within training. These results motivate adjusting target contributions, although a frequently occurring or already predictable token is not automatically uninformative. The effect of an intervention depends on the full context and the normalization of the objective.

Selective token supervision. RHO-1 [68] selects tokens using excess loss relative to a reference model trained on high-quality data. A simplified selection rule is

$$w_{s,i,t} = \mathbf{1}[\ell_{i,t}(\theta_s) - \ell_{i,t}(\theta_{\text{ref}}) \ge \gamma_s], \tag{19}$$

where γ_s can be chosen to retain a prescribed fraction of targets. The rule prioritizes positions that remain difficult for the current model but are predictable to the reference. Reference-model quality and domain coverage therefore influence which signals are retained. This changes the optimization objective, rather than merely saving storage or removing documents.

Sequence transformation for infilling. Fill-in-the-Middle (FIM) [69] instead changes the sequence transformation. For a prefix p, middle span m, and suffix u, a representative arrangement presents p and u before the target m, separated by special markers. The model learns $p_\theta(m \mid p, u)$ through causal decoding of the transformed sequence; its attention is still causal in that ordering. StarCoder [70], Code LLaMA [71], and DeepSeek-Coder-V2 [72] use this approach for code. DeepSeek V3 and DeepSeek V4 also describe infilling-related training. Their results support compatibility with left-to-right generation in the reported settings, not a guarantee for every transformation rate.

Teacher-distribution targets. Knowledge distillation changes the target distribution. With teacher probabilities p_{T} at the same prediction context, a representative token objective is

$$\ell_{i,t}^{\text{KD}} = -\sum_{v \in \mathcal{V}} p_{\text{T}}(v \mid c_{i,t}) \log p_\theta(v \mid c_{i,t}). \tag{20}$$

Gemma 2 [73] uses pretraining distillation for its 2B and 9B models. Teacher targets provide information about alternative continuations, while adding teacher computation and possible biases. They should be distinguished from sampling teacher-generated documents, which changes the corpus itself. Combining hard and soft targets introduces a further weighting choice.

Metadata as conditioning context. Metadata conditioning exposes contextual information that ordinary text extraction may discard. A document's URL, source hierarchy, or timestamp can be represented as a prefix z, yielding body-token predictions $p_\theta(x_t \mid z, x_{<t})$ with the metadata positions excluded from the loss. Allen-Zhu and Li [74] motivate studying how presentation context affects knowledge acquisition. MeCo [75] reports faster convergence using metadata prefixes, including matching a standard baseline with approximately 33% fewer training tokens in its settings. Beyond URLs [76] studies the granularity of metadata, and KoCo [77] represents metadata through composable knowledge coordinates. These approaches provide a mechanism for contextual conditioning, but their benefits depend on metadata reliability and training–inference consistency. Public evidence for transfer to frontier-scale pretraining remains limited.

V. Pretraining Evaluation

Evaluation connects data interventions to observed training outcomes. We first define the metrics used to monitor and compare recipes (§V-A), and then examine measurement and comparability limitations (§V-B).

A. Methods and Metrics

Definition 4 (Evaluation protocol and data-recipe effect)**.** *Let $\mathcal{E} = \{e_j\}_{j=1}^{n_\mathcal{E}}$ be evaluation examples and Γ specify prompting, normalization, decoding, and scoring. For a score a with larger values indicating better performance, define*

$$A(\theta; \mathcal{E}, \Gamma) = \frac{1}{n_\mathcal{E}} \sum_{j=1}^{n_\mathcal{E}} a(\theta, e_j; \Gamma). \tag{21}$$

The observed effect of recipes π and π' is $\Delta A = A(\theta_\pi; \mathcal{E}, \Gamma) - A(\theta_{\pi'}; \mathcal{E}, \Gamma)$, conditional on their training configurations and budgets.

Attributing ΔA to a data decision requires controlling other differences, including architecture, optimization, exposure, and checkpoint selection. Matching token budgets and matching compute budgets answer different questions. Repeated runs, where feasible, estimate variation due to initialization and training order; resampling evaluation examples estimates a different source of uncertainty.

Within-run diagnostics. Training loss and gradient statistics are useful within-run diagnostics. Loss spikes, plateaus, or sustained drift can indicate changes in the sampled data or numerical behavior, but do not uniquely identify their causes. PaLM describes recovery from unstable steps, while DeepSeek V3 and Kimi K2 report monitoring and stabilization procedures. Gradient and parameter norms complement the loss when diagnosing an event. A checkpoint rollback or batch exclusion should follow investigation rather than be inferred from a single universal threshold. Absolute training losses are generally unsuitable for comparisons across different tokenizers and data distributions.

Likelihood-based evaluation. Held-out negative log-likelihood provides a continuous measure of prediction quality. For fixed tokenized evaluation data, define

$$\mathcal{L}_\mathcal{E} = -\frac{1}{T_\mathcal{E}} \sum_{x \in \mathcal{E}} \sum_t \log p_\theta(x_t \mid x_{<t}), \tag{22}$$

where $T_\mathcal{E}$ counts evaluated targets. Perplexity is $\exp(\mathcal{L}_\mathcal{E})$. For a raw-text evaluation corpus, bits per byte (BPB) is

$$\mathrm{BPB}(\theta; \mathcal{E}) = \frac{-\sum_{x \in \mathcal{E}} \log p_\theta(\tau(x))}{\ln(2) \sum_{x \in \mathcal{E}} |x|_{\text{bytes}}}. \tag{23}$$

Byte normalization reduces dependence on tokenizer granularity. It does not remove differences in text normalization, context truncation, boundary handling, or the raw evaluation sample. Reported uses of byte-normalized evaluation in DeepSeek V3 and OLMo 3 should be interpreted with those protocol details fixed.

Task performance and metric sensitivity. Downstream benchmarks measure performance on tasks of interest rather than average text prediction. Table III records benchmark coverage in the source inventory. Base models are often evaluated with few-shot prompts or likelihood-based option scoring to reduce output-format ambiguity. However, exact match and accuracy discretize continuous prediction changes. The resulting curves may remain flat before a decision boundary is crossed, even when likelihood improves gradually.

Schaeffer et al. [97] show that many reported emergent patterns depend on the evaluation metric. Replacing discontinuous scoring with continuous measures can reveal smoother trends; this does not prove that all capability transitions are artifacts. Gu et al. [98] show that answer-letter and cloze-style multiple-choice protocols can provide different signals early in pretraining. The relevant variable is therefore not only the benchmark name but the full protocol Γ. Correct-answer NLL, probability-based scores, and task accuracy provide complementary observations of the same checkpoint.

Predicting performance from proxy runs. Gadre et al. [99] use correct-answer NLL to improve extrapolation from smaller runs. Their procedure first models the dependence of a continuous loss on training scale, then fits its relationship to downstream error. A representative fitted form discussed in this line of work is

$$\mathrm{Err}(L) = \varepsilon - k \exp(-\gamma L), \tag{24}$$

where the parameters and valid range are determined by the task and calibration runs. The reported small-model predictions, including the approximately 1/300 compute comparison, depend on the full experimental procedure. Equation (24) is an empirical mapping, not a universal law or a guarantee of valid extrapolation outside the fitted range. LLaMA 3's evaluation workflow illustrates the broader use of predictive experiments to inform large-scale training.

Checkpoint and stage decisions. Repeated checkpoint evaluation supports mixture and stage decisions. Phi-4 uses internal capability probes to guide data allocation and reports the importance of retaining natural web data for knowledge-oriented performance. Gemma 3 [100] combines capability benchmarks with validation perplexity when assessing model

TABLE III
BENCHMARK COVERAGE RECORDED FOR THE REPRESENTATIVE MODEL REPORTS. A CHECK DENOTES A RECORDED EVALUATION; A BLANK DENOTES NO ENTRY IN THIS INVENTORY, NOT PROOF OF NON-USE. COVERAGE ALONE DOES NOT ESTABLISH THAT CHECKPOINTS, PROMPTS, OR TRAINING STAGES ARE COMPARABLE.

Category	Benchmark	DeepSeek V4	LLaMA 3/3.1	Qwen 3	Kimi K2	GLM-5	InternLM 2	Gemma 3	Phi-4	Falcon 3	OLMo 3
Knowledge & Exam	MMLU [78]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	MMLU-Pro [79]	✓	✓	✓	✓			✓	✓	✓	
	C-Eval [80]	✓				✓	✓				
	AGIEval [81]	✓					✓	✓			
Reasoning & Common Sense	HellaSwag [82]	✓	✓			✓		✓			✓
	ARC-Challenge [83]		✓					✓		✓	✓
	WinoGrande [84]	✓	✓					✓		✓	✓
	BBH [85]	✓	✓	✓	✓	✓		✓		✓	
	GPQA (Diamond) [86]		✓	✓	✓			✓	✓	✓	
	PIQA [87]							✓		✓	✓
Mathematics	GSM8K [88]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	MATH [89]	✓	✓	✓	✓		✓	✓	✓	✓	
	MGSM [90]	✓		✓				✓			
Code Generation	HumanEval [91]	✓	✓		✓	✓	✓	✓	✓		
	MBPP [92]		✓		✓		✓	✓	✓		
	EvalPlus [93]		✓	✓							
Reading & QA	TriviaQA [94]	✓					✓	✓			✓
	NaturalQuestions [95]							✓			✓
	DROP [96]	✓						✓			✓

choices. A useful summary retains a vector of task scores rather than only a single aggregate, because domain-specific improvements can coexist with regressions elsewhere. Figure 5 separates this development feedback from final held-out assessment.

Evaluation also informs when to end a training stage. LLaMA 3 reports 24.0- and 6.4-percentage-point improvements on GSM8K and MATH for the 8B model during annealing, with smaller gains for the 405B model. A limited gain at a particular checkpoint indicates that the tested continuation is less beneficial under that configuration; it does not prove that the model has extracted all useful information from the corpus. OLMo 2 separates development tasks from held-out tasks, and OLMo 3 broadens its benchmark coverage. Such practices support stopping decisions based on marginal gain, cost, and uncertainty rather than a claim of complete capability saturation.

Resource-normalized comparisons. An operational efficiency measure can be defined for a fixed target score a_0 and protocol as

$$F_\pi(a_0) = \inf\{F : A(\theta_{\pi,F}; \mathcal{E}, \Gamma) \geq a_0\}. \tag{25}$$

A speedup $F_{\pi'}(a_0)/F_\pi(a_0)$ is meaningful only if both recipes reach that target under comparable measurement. An analogous definition uses exposure B for data efficiency. When relevant, end-to-end comparisons should additionally report corpus processing, teacher inference, synthesis, and search costs. This makes explicit which resource is saved by a data intervention.

B. Measurement Challenges

Benchmark scores are proxies for capabilities, not complete descriptions of them. A change in A can reflect better knowledge, adaptation to a format, contamination, or a trade-off between domains. Consequently, an association between a recipe and a score is insufficient to establish a causal mechanism. Controlled comparisons and multiple complementary measurements reduce ambiguity without eliminating it.

Transfer to downstream adaptation. One limitation concerns transfer beyond the base-model evaluation setting. Agentic applications require tool interaction, planning, and environmental feedback that standard text benchmarks do not directly assess. Yi-Lightning [101] reports divergence between preference-based rankings and conventional benchmark performance. These are post-training observations and are relevant here as evidence about the limits of evaluation transfer, not as direct evidence for a pretraining data recipe. Ministral 3 [102] similarly reports that teacher rankings observed during pretraining distillation can differ from those after post-training. A base-model score alone may therefore be insufficient to predict the value of a checkpoint for later adaptation.

Confounded capability measurements. Another limitation is capability confounding. MMLU requires knowledge, contextual interpretation, and compliance with an answer format. A change in its aggregate score does not identify which component improved. FineWeb-Edu reports gains on some educational or knowledge-oriented tasks alongside weaker results on other commonsense evaluations. Such outcomes are consistent with a shift in the retained distribution rather than a uniform improvement in every capability. Reporting the vector of task-level effects, together with domain retention statistics, makes these trade-offs more visible.

Protocol-dependent comparisons. Cross-experiment comparability adds further uncertainty. Lourie et al. [103] find that only a subset of the downstream tasks examined in the scaling setup of Gadre et al. [99] display monotonic trends. Prompt

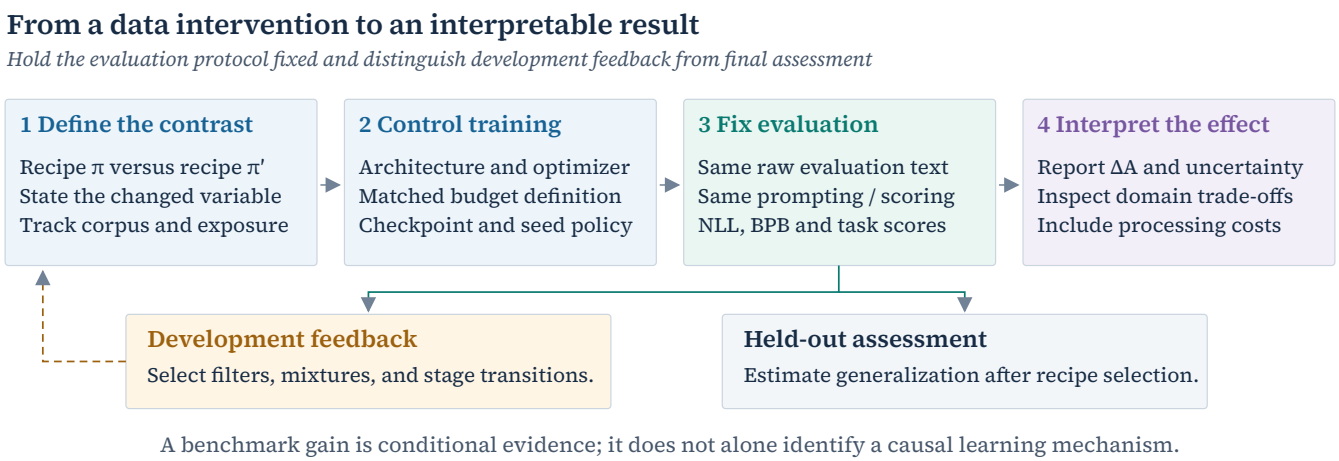


Fig. 5. Evaluation of a data intervention. Controlled training and a fixed evaluation protocol connect recipe changes to observed effects. Development feedback is separated from held-out assessment to limit adaptive overfitting.

format, demonstration count, and the validation distribution can change both trends and data rankings. FineWeb-Edu and DCLM also use different model scales, training budgets, and evaluation protocols in their headline results. Those figures cannot establish a common ranking of corpora without an aligned experiment. The color groups in Table II preserve the distinction between evaluation settings rather than combining their scores.

Contamination and adaptive overfitting. Evaluation contamination and adaptive selection create separate risks. Text overlap between $\mathcal{D}$ and $\mathcal{E}$ may expose benchmark material during training, while repeated optimization on a development benchmark can overfit its preferences even without direct overlap. Exact and fuzzy decontamination should therefore be documented independently of ordinary corpus deduplication. Held-out evaluation should remain separate from the feedback used to choose filters, mixtures, and stage transitions. These controls support interpretation of ΔA, but a lack of detected lexical overlap does not prove that all benchmark-related information was excluded.

Controlled evaluation of mechanisms. Controlled synthetic evaluations can help isolate mechanisms that aggregate benchmarks confound. The Physics of Language Models studies examine knowledge storage and extraction [74], knowledge manipulation [104], and the relation between parameter count and knowledge capacity [105]. By controlling how facts and tasks are generated, such experiments can attribute effects more precisely within their designed setting. They complement natural-data benchmarks rather than replace them. Whether a mechanism identified in a controlled synthetic world persists under heterogeneous frontier pretraining remains an empirical question.

VI. Trends in Pretraining Data Engineering

This section synthesizes the preceding evidence into a shared pipeline (§VI-A) and four recurring technical themes (§VI-B). Each theme separates the supported observation, its practical implication, and the unresolved boundary of the evidence.

A. A Shared Pipeline for Pretraining Data

Pipeline components and feedback. Figure 6 relates the three stages reviewed in this survey. Corpus construction determines the eligible examples $\mathcal{D} = \mathcal{C}(\mathcal{R})$. Utilization determines their presentation through $\pi_s = (q_s, \mathcal{T}_s, L_s, w_s)$ and interacts with the optimization schedule. Evaluation measures outcomes through $A(\theta; \mathcal{E}, \Gamma)$ and provides feedback for subsequent decisions. The common structure does not imply that every report uses the same algorithms or that all components have independent measured benefits.

Interpreting reported adoption. Table IV summarizes the model inventory using conservative reporting codes. An unknown entry is not evidence of absence, and synthetic post-training or multimodal processing is not automatically counted as synthetic text pretraining. A learning-rate decay alone is not coded as quality-focused annealing. Token budgets retain the reviewed sources' reporting scope: some describe repeated exposure or multimodal training, so the table should not be read as a ranking of unique text supply. In particular, Kimi K2.5 and LLaMA 4 provide broader model context rather than matched text-only experiments. Appendix A retains the field-level evidence and its limitations.

B. Technical Focus and Trends

We use *consensus* to denote recurring observations across the reviewed reports, not a universal optimum or a causal claim established by a meta-analysis. The four themes below retain the progression from scale and efficiency to synthesis, scheduling, and transparency.

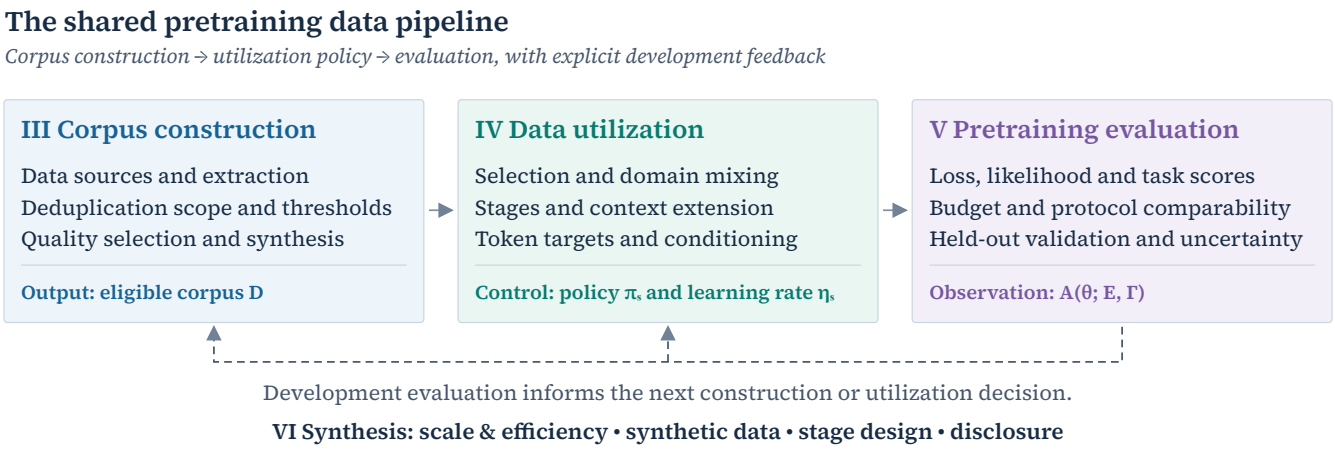


Fig. 6. Shared organization of pretraining data engineering. The pipeline follows Sections III–V; the feedback path represents development decisions rather than access to held-out test answers.

Consensus 1. Scale and data efficiency are complementary

Supported observation. Filtering, deduplication, and data allocation affect performance at a fixed training budget, while larger token budgets remain part of reported frontier recipes.

Practical implication. Report corpus size U, exposure B, and compute F separately, and compare efficiency at a common evaluation target.

Open boundary. Gains from a smaller or more selective corpus need not persist at larger model scales, longer training horizons, or new capability targets.

Compute-optimal scaling describes a particular resource allocation problem [2]. Later work considers additional deployment and inference constraints that can favor training beyond that allocation [99, 106]. RefinedWeb, FineWeb-Edu, and DCLM demonstrate that corpus processing can change performance under controlled budgets, but their different protocols do not yield a single transferable multiplier. Equation (25) makes the conditional nature of a speedup explicit. The evidence supports joint optimization of scale and data use rather than a general replacement of one by the other.

Consensus 2. Synthetic data has a conditional, heterogeneous role

Supported observation. Phi, Qwen, and Kimi report generation or rewriting, with different objectives and exposure shares. Disclosure is insufficient to infer a universal adoption rate.

Practical implication. Distinguish source-grounded rewriting from new generation and state the stage, validation procedure, and mixture convention.

Open boundary. No universal value of ρ in Eq. (13) is established across domains, model scales, and repeated synthesis rounds.

Phi emphasizes textbook-like material, Qwen uses domain-specific generation, and Kimi describes source-conditioned transformations. These are distinct uses of models in corpus construction rather than interchangeable implementations of one recipe. Reports that emphasize natural sources, including DeepSeek V3, do not establish exhaustive absence of synthetic content. Model-collapse studies identify degradation under specific recursive generation and replacement conditions [48, 107]. Their implications for mixed natural–synthetic frontier training depend on data retention, sampling, and verification, so neither universal collapse nor universal improvement follows from the presence of synthetic data alone.

Consensus 3. Staged training is established; stage design remains empirical

Supported observation. Reports increasingly describe changes in mixtures and context lengths across training, sometimes alongside learning-rate decay.

Practical implication. Specify stage boundaries and changes to q_s, $\mathcal{T}_s$, L_s, w_s, and η_s separately.

Open boundary. Optimal transitions and proxy-to-target transfer remain conditional on scale, budget, and the desired capability profile.

LLaMA 1's fixed mixture, LLaMA 3's iterative mixture refinement and annealing, and Qwen 3's distinct stages illustrate increasingly explicit data scheduling. DeepSeek V3, Kimi K2, and GLM-5 provide additional examples of staged training, with different disclosure granularity. DoReMi, RegMix, and Data Mixing Laws formalize parts of the allocation problem [51–53], while AutoScale documents transfer limitations [57]. These methods inform Eq. (17), but do not yet determine universal stage boundaries. Capability-focused staging should also remain distinct from a globally ordered difficulty curriculum.

Consensus 4. Disclosure constrains reproducibility and inference

Supported observation. Model weights and architecture can be available even when corpus sources, mixture weights, and processing choices remain undisclosed.

Practical implication. Record source provenance, stage-specific claims, and disclosure level; preserve unknowns rather than converting them into negative evidence.

Open boundary. Incomplete reporting limits causal comparison and does not, by itself, establish that data engineering explains all performance differences.

TABLE IV
REPORTED PRACTICES IN THE MODEL INVENTORY. ✓ = EXPLICIT DESCRIPTION IN THE REVIEWED EVIDENCE; △ = PARTIAL OR INDIRECT EVIDENCE; ? = INSUFFICIENT PRETRAINING-SPECIFIC DETAIL. DOC/SPAN = DOCUMENT/SUBDOCUMENT DEDUPLICATION; RULE/MODEL = FILTERING; SYN. = SYNTHETIC TEXT; MIX = OPTIMIZED MIXTURE; STAGES = STAGED TRAINING; Q-ANN. = QUALITY-FOCUSED ANNEALING. THESE CODES DESCRIBE REPORTING, NOT CAUSAL EFFECTIVENESS.

			Dedup		**Filter**			**Strategy**		
Model	**Date**	**Tokens**	Doc	Span	Rule	Model	**Syn.**	Mix	Stages	Q-Ann.
GLM-130B	2022.10	0.4T	?	?	✓	?	?	?	?	?
BLOOM	2022.07	0.35T	✓	?	✓	?	?	?	?	?
LLaMA	2023.02	1.4T	✓	✓	✓	✓	?	?	?	?
Falcon	2023.05	1.5T	✓	△	✓	✓	?	?	?	?
LLaMA 2	2023.07	2T	△	?	△	?	?	?	?	?
Qwen	2023.09	3T	✓	?	✓	✓	?	?	?	?
Phi-2	2023.12	1.4T	?	?	△	△	✓	?	?	?
MiniCPM	2024.04	1T	△	?	△	△	?	?	✓	✓
DeepSeek V2	2024.05	8.1T	✓	?	✓	✓	?	?	△	?
Gemma 2	2024.06	13T	✓	?	✓	✓	?	△	?	?
LLaMA 3	2024.07	15T+	✓	✓	✓	✓	△	✓	✓	✓
Qwen 2.5	2024.09	18T	✓	?	✓	✓	✓	✓	✓	△
DeepSeek V3	2024.12	14.8T	✓	?	✓	✓	?	✓	✓	?
Phi-4	2024.12	10T	△	?	△	△	✓	?	✓	✓
InternLM 3	2025.01	4T	△	?	△	△	△	△	△	△
OLMo 2	2025.01	4T	✓	?	✓	✓	△	△	✓	✓
Gemma 3	2025.03	14T	△	?	△	△	?	?	△	?
LLaMA 4	2025.04	40T	?	?	△	?	△	?	△	△
Qwen 3	2025.05	36T	?	?	△	✓	✓	✓	✓	△
OLMo 3	2025	5.9T	✓	?	✓	✓	✓	✓	✓	✓
Kimi K2	2025.07	15.5T	△	?	△	✓	✓	△	✓	?
GLM-4.5	2025.07	23T	△	?	△	✓	△	△	✓	?
Kimi K2.5	2026.02	15T	?	?	△	?	?	△	△	?
GLM-5	2026.02	28.5T	△	?	△	✓	△	△	✓	?
DeepSeek V4	2026.04	32T	△	?	△	△	?	△	✓	?

The reports differ substantially in what they disclose. LLaMA 1 specifies source proportions, while the LLaMA 4 release provides less methodological detail in the reviewed sources [108]. Kimi K2 describes aspects of synthesis without a complete corpus distribution. Mistral 7B [109] and Mixtral [23] disclose limited pretraining-data information. Conversely, OLMo and open corpus projects provide data and processing artifacts that support reproduction. This unevenness supports field-level evidence coding, but does not prove a monotonic relation between capability, commercial value, and secrecy. Recurring practices should therefore be inferred from corroborated disclosures, with disagreements and unknowns retained explicitly.

VII. CONCLUSION AND FUTURE DIRECTIONS

This survey has examined data engineering for text-based LLM pretraining through the progression from corpus construction to utilization and evaluation. Public technical reports and open corpus projects reveal recurring practices, including duplicate control, layered quality filtering, domain-aware sampling, and staged training. Their reported effectiveness remains conditional on model scale, data availability, compute, and evaluation protocol. A shared pipeline is therefore useful for organizing evidence, but should not be interpreted as a universally optimal recipe.

The formalization developed here treats data as a set of coupled decisions. Corpus construction determines $\mathcal{D}$, while the utilization policy π_s controls exposure, context, and target weighting. Evaluation under a fixed protocol connects these choices to measured outcomes. This perspective clarifies why additional pipeline complexity need not improve performance: a selective filter can reduce useful coverage, a proxy optimum can fail to transfer, and a benchmark gain can conceal regressions elsewhere. Broad source coverage and controlled interventions remain necessary complements to targeted optimization.

An important direction is to understand how presentation context affects knowledge acquisition. Standard preprocessing often removes provenance and contextual metadata before training, although these fields may help distinguish the conditions under which a statement is produced and used. Our ongoing work on knowledge context[1] investigates how such information can inform pretraining data engineering. The central questions concern which context is useful, how it should be represented, and whether improvements persist across domains and scales. These questions require controlled evaluation rather than an assumption that adding metadata is always beneficial.

Further progress also depends on reliable evidence from actual training. Releasing corpus statistics, processing configurations, stage-level mixtures, and evaluation protocols would make reported gains easier to interpret and reproduce. Distinguishing training exposure from stored volume, and reported adoption from demonstrated effectiveness, is especially important as natural and synthetic data become more closely integrated. This survey provides a basis for organizing that evidence and for revising the resulting empirical consensus as additional reports and open artifacts become available.

[1] https://ydli-ai.github.io/#knowledge-context

APPENDIX A
PRETRAINING DATA DISCLOSURE EVIDENCE

Coding scheme. L1 denotes mentioned-only disclosure; L2 denotes qualitative source or method description; L3 denotes quantitative disclosure such as token counts, proportions, filtering rates, or stage sizes; and L4 denotes released data/code/recipes or enough detail for approximate reproduction. Blank cells in the source survey are shown as –. Evidence snippets intentionally stay close to the cited technical report or model card.

Scope and interpretation. This appendix presents an inventory of reports, release notes, and model cards. Some entries concern multimodal training, distillation, supervised fine-tuning, or reinforcement learning. Such snippets remain contextual disclosure evidence and are not counted as confirmation of text-only pretraining practices in Section VI. In particular, a field headed "Mid" may reproduce a source's broader terminology rather than the pretraining definition used in the main text. "Syn." includes heterogeneous synthetic or distilled signals and does not always mean generated pretraining documents. A dash denotes missing or insufficient evidence, not confirmed non-use. Disclosure levels describe the specificity of reporting, not methodological effectiveness or independent verification. The same report may supply evidence for multiple model versions.

Inventory summary. The table contains 54 model entries. Its recorded peak codes comprise 5 at L4, 32 at L3, 11 at L2, 1 at L1, and 5 without an assigned level. These are inventory counts, not adoption rates for text pretraining. Source locators are retained for follow-up verification, including entries whose available bibliographic record is incomplete.

TABLE V: Evidence table for public disclosure of pretraining data in LLM technical reports and model cards. Each cell records the highest available evidence fragment for the corresponding data-engineering aspect.

Model	Org. / Scope	Peak	Corpus Sources	Processing / Dedup.	Mixture and Training Schedule	Synthetic / Eval Evidence	Ref.
BLOOM 2022-07	BigScience Multilingual	L4	L4 ROOTS; 46 natural + 13 programming languages (Sec. 3.1/T1)	L4 ROOTS pipeline; tools released (Sec. 3.1/F2)	Mix: L4 language/source bytes in T1 (Sec. 3.1/T1)	Eval: L4 eval code + xP3/ ROOTS docs (Sec. 5)	[110]
GLM-130B 2022-10	Tsinghua / Zhipu Bilingual	L3	L3 Pile 1.2T; WuDao 1.0T; corpora total 400B tokens (Sec. 3.1)	L1 filtering pipeline named (Sec. 3.1)	Mix: L3 source token counts (Sec. 3.1)	Syn.: L3 MIP 5% tokens; T0++/DeepStruct (Sec. 3.1)	[111]
LLaMA 1 2023-02	Meta General	L3	L3 CC/C4/GitHub/Wikipedia/books/ arXiv/StackExchange (Sec. 2.2/T1)	L3 CCNet; Wikipedia classifier; near-dedup (Sec. 2.2)	Mix: L3 CC 67%; C4 15%; GitHub 4.5%; Wikipedia 4.5%; Books 4.5%; arXiv 2.5%; StackExchange 2% (T1)	–	[112]
ChatGLM-6B 2023-03	Tsinghua / Zhipu Bilingual chat	L3	L3 ∼1T Chinese-English tokens (README/model card)	–	Mix: L1 Chinese + English corpus	–	[113]
Pythia 2023-04	EleutherAI Research / general	L4	L4 The Pile + Pile-dedup (Sec. 3)	L4 data order + dataloaders reconstructable (Sec. 3)	Mix: L4 fixed order; all checkpoints released (Sec. 3)	Eval: L4 training/eval code released (Sec. 3/Sec. 4)	[114]
Falcon / RefinedWeb 2023-05	TII English / general	L4	L3 CommonCrawl-derived RefinedWeb; 5T web tokens (Sec. 3)	L4 MDR; URL filter; fastText; MinHash + suffix-array dedup (Sec. 3/T2/F2)	Mix: L2 web-only design (Sec. 3)	–	[115]
Baichuan 1 2023-06	Baichuan Chinese-English	L3	L3 7B 1.2T; 13B 1.4T tokens (report/ model card)	L1 cleaning/filtering named	Mix: L1 Chinese + English	–	[116]
ChatGLM2-6B 2023-06	Tsinghua / Zhipu Bilingual chat	L3	L3 1.4T bilingual tokens (README/ model card)	–	Mix: L1 bilingual corpus	–	[117]
Phi-1 2023-06	Microsoft Code	L3	L3 textbook-quality code corpus; token counts (Sec. 2)	L2 textbook-quality filtering (Sec. 2)	Mix: L3 token-scale mixture (Sec. 2)	Syn.: L3 synthetic textbooks/ exercises (Sec. 2)	[118]
LLaMA 2 2023-07	Meta General	L3	L3 public data; 2T tokens (Sec. 2.2)	L1 safety filters; personal-data filtering (Sec. 2.2/Sec. 4)	Mix: L1 factual-source upsampling (Sec. 2.2)	–	[119]
Baichuan 2 2023-09	Baichuan Multilingual / Chinese-English	L3	L3 2.6T tokens (Abstract/Sec. 1)	L1 safety/filtering named (Sec. 6)	Mix: L1 multilingual + law/medical focus (Sec. 1)	–	[120]
InternLM 1 2023-09	Shanghai AI Lab Bilingual / general	L3	L3 1.6T tokens; web/books/code/ academic/encyclopedia (Sec. 2.1/T1)	L3 language ID; rules; MinHash; safety (Sec. 2.1)	Mix: L2 multi-stage data recipe (Sec. 2.1) Mid: L2 capability-stage data (Sec. 2.1)	–	[121]
Phi-1.5 2023-09	Microsoft Reasoning / small LM	L3	L3 phi-1 corpus + synthetic textbooks (Sec. 2)	L2 quality-first corpus (Sec. 2)	Mix: L3 7B code + 20B synthetic (Sec. 2)	Syn.: L3 ∼20B synthetic textbook-like tokens (Sec. 2)	[122]

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TABLE V: Evidence table for public disclosure of pretraining data in LLM technical reports and model cards (continued).

Model	Org. / Scope	Peak	Corpus Sources	Processing / Dedup.	Mixture and Training Schedule	Synthetic / Eval Evidence	Ref.
Qwen 1 2023-09	Alibaba Bilingual / multilingual	L3	L3 ∼3T tokens; text + code (Sec. 2.1)	L3 rules; ML scoring; exact + MinHash dedup (Sec. 2.1)	Mix: L2 source/category mix (Sec. 2.1)	–	[123]
ChatGLM3-6B 2023-10	Tsinghua / Zhipu Bilingual chat / tools	L1	L1 third-generation pre-trained dialogue model (README/model card)	–	Mid: L1 tools/function-calling training data	–	[124]
Mistral 7B 2023-10	Mistral AI General	–	–	–	–	–	[125]
Grok-1 2023-11	xAI General	–	–	–	–	–	[126]
Mixtral 8x7B 2023-12	Mistral AI General / MoE	–	–	–	–	–	[127]
Phi-2 2023-12	Microsoft Reasoning / small LM	L3	L3 synthetic + filtered web data; token scale (model card)	L2 quality filters (model card)	Mix: L2 curated blend (model card)	Syn.: L3 synthetic data central (model card)	[128]
Gemma 1 2024-02	Google English / code	L3	L3 2T/6T tokens; web/code/math (Sec. 2)	L2 unsafe/PII/recitation filters (Sec. 4/Sec. 8)	Mix: L1 web + code + math categories (Sec. 2)	Syn.: L1 post-train synthetic prompts/responses (Sec. 4)	[129]
OLMo 1 2024-02	AI2 General / open science	L4	L4 Dolma; CC/Stack/C4/Reddit/ peS2o/books/Wikipedia (Sec. 2.2/T3)	L4 data + curation tools + logs released (Sec. 2.2/Sec. 5)	Mix: L4 Dolma source tokens in T3 (Sec. 2.2/T3)	Eval: L4 OLMo-Eval + Catwalk (Sec. 2.4/Sec. 5)	[34]
DBRX 2024-03	Databricks General / MoE	L2	L2 12T tokens; web/code/math/NL (report/blog)	L1 curated data pipeline	Mix: L2 curriculum/mix categories Mid: L1 continued training stage	Syn.: L2 synthetic post-training/instruction data	[130]
InternLM 2 2024-03	Shanghai AI Lab Bilingual / general	L3	L3 ∼2T+ tokens; web/code/books/ knowledge (Sec. 2.1/T2)	L3 quality/safety/dedup pipeline (Sec. 2.1)	Mix: L2 multi-stage curriculum (Sec. 2.1) Mid: L2 capability + long-context stages (Sec. 2)	Syn.: L2 knowledge/ capability augmentation (Sec. 2.1)	[36]
Yi / Yi-1.5 2024-03	01.AI Bilingual	L3	L3 3.1T English-Chinese tokens (Abstract/Sec. 2)	L3 heuristic + learned + cluster filters; MinHash (Sec. 2.1)	Mix: L3 bilingual/source mixture figures (Sec. 2.1/F2) Mid: L2 10B long-context continuation (Sec. 2.3)	–	[41]
Command R / R+ 2024-04	Cohere Multilingual / RAG	L2	L1 multilingual + retrieval data (model card)	L1 filtering named	Mix: L2 RAG/tool-use mixture Mid: L1 continued training	Syn.: L2 tool/RAG post-training data	[131]
LLaMA 3 2024-04	Meta General / multilingual	L2	L2 15T+ tokens; 4× code; >5% non-English (blog)	L2 HTML extraction; dedup; quality classifiers (blog)	Mix: L2 scaling-law data mix (blog) Ann.: L2 annealing/cooldown data (report)	Syn.: L1 Llama 2 labels for quality classifiers (blog)	[132, 133]
MiniCPM 2024-04	ModelBest / OpenBMB Bilingual / small LM	L3	L3 CC-Chn/Dolma/C4/Pile/Stack/ StarCoder (Sec. 6.3/F11)	L3 MinHash within + across corpora (Sec. 6.3)	Mix: L3 stage data distributions (F11) Mid: L2 WSD two-stage schedule (Sec. 5/ Sec. 6.2)	Syn.: L2 UltraChat/SlimOrca/ OssInstruct + proprietary SFT (Sec. 6.3)	[134]
Phi-3 2024-04	Microsoft Small LM / multilingual	L3	L3 3.3T/4.8T tokens; web + synthetic educational data (Sec. 3)	L2 heavy filtering (Sec. 3)	Mix: L2 two-phase data blend (Sec. 3) Mid: L2 long-context continuation (Sec. 3)	Syn.: L3 synthetic data core (Sec. 3)	[135]
MAP-Neo 2024-05	MAP Bilingual / open science	L4	L4 Matrix; 4.5T tokens; corpus released (Abstract/website)	L4 cleaned corpus + cleaning pipeline released (Abstract)	Mix: L3 CC 52.55%; code 22.29%; papers/books/etc. (website) Mid: L2 two-stage pre-training (website)	–	[136]
GLM-4 2024-06	Zhipu AI Multilingual / tools	L2	L2 webpages/Wikipedia/books/code/ papers (Sec. 2)	L2 exact/fuzzy dedup; filters (Sec. 2)	Mix: L1 multilingual mixture (Sec. 2) Mid: L1 tool-capability training (Sec. 3)	Syn.: L1 reasoning/tool data (Sec. 3)	[137]
Gemma 2 2024-06	Google English / general	L3	L3 2T/8T/13T tokens; web/code/ science (Sec. 3.1)	L2 unsafe/PII/decontam/ recitation filters (Sec. 3.1)	Mix: L2 mixture chosen by ablations (Sec. 3.1)	Syn.: L1 distillation targets for 2B/9B (Sec. 3.2) Eval: L2 eval-set decontamination (Sec. 3.1)	[138]
Nemotron 4 2024-06	NVIDIA General / synthetic-data generation	L3	L3 9T tokens (Introduction)	L2 high-quality dataset referenced (Intro)	Mix: L2 post-train SFT/RL domains (Sec. 3) Mid: L2 continued training (Sec. 2.3)	Syn.: L3 post-train >98% synthetic (Abstract/Sec. 1)	[139]

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TABLE V: Evidence table for public disclosure of pretraining data in LLM technical reports and model cards (continued).

Model	Org. / Scope	Peak	Corpus Sources	Processing / Dedup.	Mixture and Training Schedule	Synthetic / Eval Evidence	Ref.
Qwen 2 2024-06	Alibaba Multilingual	L3	L3 7T tokens; 27+ languages (Sec. 2)	L2 model-assisted filtering + dedup (Sec. 2)	Mix: L2 multilingual/code data (Sec. 2) Mid: L2 long-context/capability stages (Sec. 2)	Syn.: L2 synthetic high-quality data (Sec. 2)	[140]
LLaMA 3.1 2024-07	Meta Multilingual / long context	L2	L2 Llama 3 continuation; multilingual + long-context data (Sec. 3)	L2 Llama 3 pipeline inherited (Sec. 3)	Mix: L2 long-context mix (Sec. 3) Mid: L2 128K continued pre-training (Sec. 3) Ann.: L2 cooldown data (Sec. 3)	Syn.: L1 post-training/eval synthetic data (Sec. 4)	[132]
Qwen 2.5 2024-09	Alibaba Multilingual / code / math	L3	L3 18T tokens; math/code/ multilingual expansion (Sec. 2)	L3 filtering + decontamination in subreports (Sec. 2)	Mix: L3 domain stages/ratios in subreports Mid: L3 math/code continued training Ann.: L2 annealing stage	Syn.: L3 math/code synthetic data (Sec. 2; Coder/Math reports)	[141–143]
DeepSeek V3 2024-12	DeepSeek General / MoE	L3	L3 14.8T tokens; web pages + e-books; math/code (Sec. 2.1)	L2 preprocessing inherited; sparse details (Sec. 2.1)	Mix: L2 math/code + FIM emphasis (Sec. 2.1) Mid: L2 context extension + MTP (Sec. 2)	Syn.: L1 no intentional synthetic pre-training (transparency disclosure)	[144]
Phi-4 2024-12	Microsoft Reasoning / small LM	L3	L3 web/books/code/synthetic; token/ proportion facts (Sec. 3)	L2 filtering + quality selection (Sec. 3)	Mix: L2 web/books/code/synthetic blend (Sec. 3) Mid: L2 mid-training stage (Sec. 3)	Syn.: L3 ∼40% pre-training synthetic (Sec. 3)	[145]
Kimi k1.5 2025-01	Moonshot AI Multimodal reasoning	L2	L2 math/reasoning; knowledge; multimodal categories (App. B)	L2 FastText; OCR quality; LM filters; dedup (App. B)	Mix: L2 category-level corpus (App. B) Mid: L2 RL + long-context stages (Sec. 2)	Syn.: L2 synthetic multimodal/RL data (App. B/ Sec. 2)	[146]
MiniMax-01 2025-01	MiniMax General / long context / multimodal	L2	L2 academic literature/books/web/ code (Sec. 4.1.1)	L2 rule cleaning; dedup; reward-model quality scores (Sec. 4.1.1)	Mix: L2 balanced sampling by quality/ category (Sec. 4.1.1) Mid: L2 long-context pre-training (Sec. 4.3)	–	[147]
Phi-4-Mini 2025-02	Microsoft Small LM / multilingual	L2	L2 multilingual/code/math data (Sec. 3)	L2 filtering (Sec. 3)	Mix: L2 qualitative mixture (Sec. 3) Mid: L2 long-context/multimodal variants (Sec. 3)	Syn.: L2 synthetic data (Sec. 3)	[148]
Gemma 3 2025-03	Google Multimodal / multilingual	L2	L2 text/image/multilingual categories (Sec. 3)	L2 quality + safety filters (Sec. 3/Sec. 8)	Mix: L2 modality/language mixture (Sec. 3) Mid: L2 multimodal continued training (Sec. 3)	Syn.: L1 post-training synthetic data (Sec. 4)	[149]
LLaMA 4 Scout / Maverick 2025-04	Meta Multimodal / MoE	L2	L2 public + licensed + Meta-product data (model card)	L1 filtering/safety (model card)	Mid: L2 multimodal + long-context training (model card) Ann.: L1 cooldown/annealing (model card)	Syn.: L1 distillation/synthetic data (model card)	[150]
Qwen 3 2025-05	Alibaba Multilingual / reasoning	L3	L3 S1 >30T; S2 +5T; S3 long-context data (blog)	L3 VL-OCR; filtering; data selection (blog)	Mix: L3 S1/S2/S3 stages (blog) Mid: L3 reasoning + long-context stages (blog) Ann.: L2 annealing stage	Syn.: L3 math/code/reasoning synthetic data (blog)	[151]
MiniMax-M1 2025-06	MiniMax Reasoning / long context / RL	L2	L1 based on MiniMax-Text-01; no new base corpus (Abstract)	–	Mix: L2 diverse RL problems (Abstract) Mid: L2 CISPO RL; 40K/80K thinking budgets (Abstract)	Syn.: L2 RL on sandbox + real-world SWE environments (Abstract)	[152]
GLM-4.5 2025-07	Zhipu AI (Z.AI) Multilingual / reasoning / MoE	L3	L3 23T tokens (paper/abstract)	L2 filtering/scoring pipeline (Sec. data)	Mix: L2 pretrain + reasoning/code stages Mid: L2 agent/reasoning mid-training	Syn.: L2 reasoning synthetic data (Sec. data)	[153]
Kimi K2 2025-07	Moonshot AI Agentic / MoE	L3	L3 15.5T tokens; Web Text/Code/ Math/Knowledge (Sec. 2.2)	L2 K1.5 pipelines; correctness/ quality validation (Sec. 2.2)	Mix: L2 four pretrain domains (Sec. 2.2) Mid: L2 multi-stage post-training (Sec. 3) Ann.: L2 MuonClip; zero loss spike; LR schedule (Sec. 2)	Syn.: L3 post-train agentic data synthesis (Sec. 3)	[154]
DeepSeek V3.1 2025-08	DeepSeek General / long context / agent	L3	L3 840B-token continued pre-training (release note)	–	Mid: L2 long-context continuation (release note)	–	[155]
MiniMax-M2 2025-10	MiniMax Coding / agentic MoE	–	–	–	–	–	[156]

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TABLE V: Evidence table for public disclosure of pretraining data in LLM technical reports and model cards (continued).

Model	Org. / Scope	Peak	Corpus Sources	Processing / Dedup.	Mixture and Training Schedule	Synthetic / Eval Evidence	Ref.
DeepSeek V3.2 2025-12	DeepSeek Reasoning / agent / sparse attention	L3	L1 V3.2-Exp/V3 lineage; no full base corpus (Sec. 1)	–	Mix: L2 reasoning + tool-use post-training (Sec. post-train) Mid: L2 scalable RL + agent synthesis (Sec. post-train)	Syn.: L3 1,800+ envs; 85k+ agent instructions (Sec. post-train)	[157]
GLM-5 2026-02	Zhipu AI (Z.AI) Multilingual / MoE / agent	L3	L3 28.5T tokens (Sec. 1/abstract)	L2 data filtering/scoring (Sec. data)	Mix: L2 ∼27T pretrain + mid-training (report) Mid: L2 long-context + agentic mid-training	Syn.: L2 reasoning + agent data (Sec. data)	[158]
Kimi K2.5 2026-02	Moonshot AI Multimodal / agentic	L3	L3 ∼15T mixed visual + text tokens (Sec. 4.3/T3)	L1 inherits K2/K1.5 pipelines (Sec. 4.3)	Mix: L3 ViT + joint pretrain + mid-training stages (Sec. 4.3/T3) Mid: L2 high-quality data + long-context activation (Sec. 4.3)	–	[159]
Qwen 3.5 / Qwen3.5-Omni 2026-02	Alibaba Native multimodal / multilingual	L3	L3 Omni S2 ∼4T tokens; text 0.92T, audio 1.99T, image 0.95T, video 0.14T, video-audio 0.29T (Sec. 3)	L1 stricter filtering (blog)	Mix: L3 modality token breakdown (Sec. 3) Mid: L2 S1 encoder alignment; S2 general; S3 long-context to 262K (Sec. 3)	–	[160, 161]
DeepSeek V4 2026-04	DeepSeek Long-context / reasoning / MoE	L3	L3 V4-Pro/Flash >32T tokens; 1M context (HF card)	L2 web filtering; V3 preprocessing; sample-level attention masking (report/HF card)	Mix: L2 math/code/web/long documents (report) Mid: L2 domain experts via SFT+GRPO; on-policy distillation (HF card) Ann.: L2 LR + long-context schedule (report)	Syn.: L2 agentic data in mid-training (report)	[8]
Kimi K2.6 2026-04	Moonshot AI Multimodal / coding / agentic	–	–	–	–	–	[162]

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